

Community Emergency Response App (CERA)

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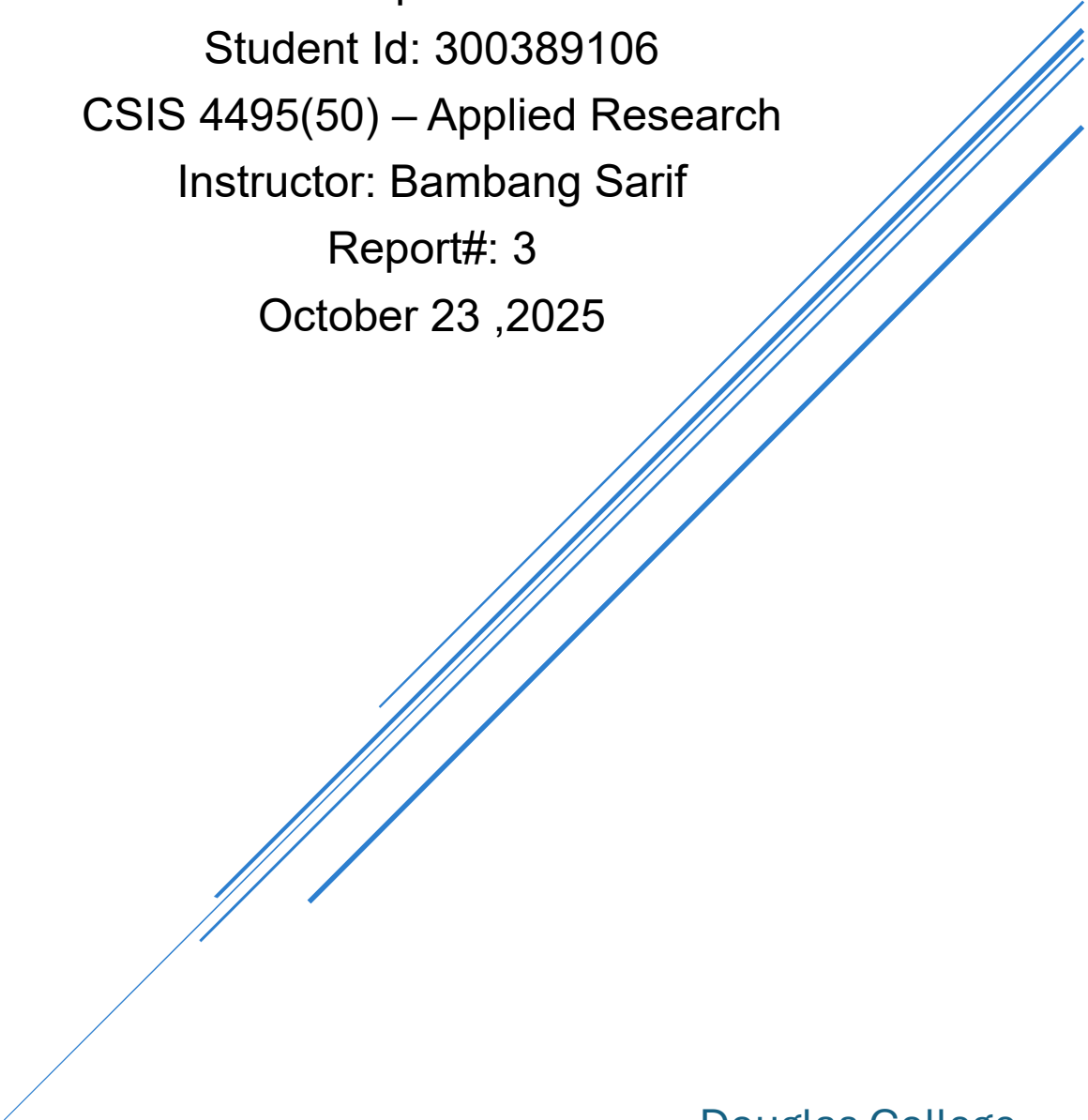
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CSIS 4495(50) – Applied Research

Instructor: Bambang Sarif

Report#: 3

October 23 ,2025

A series of five parallel blue lines of varying lengths, slanted diagonally from the bottom-left towards the top-right, positioned to the right of the student information.

Douglas College
CSIS 4495

Work Logs

Date	Number of Hours	Description of work done
Oct 14, 2025	3	updated incident schema and added (assignedVolunteer) structure with status tracking (pending, accepted, declined, in_progress, completed). Tested dispatch and assignment flow in backend.
Oct 15, 2025	5	Integrated verifyToken middleware and implemented role-based access for coordinator routes. Resolved JWT decoding and ObjectId issues.
Oct 16, 2025	6	Built complete assignment API (POST /incidents/:id/dispatch, GET /incidents/assigned/me) and linked it with frontend task screen.
Oct 18, 2025	4.5	Integrated volunteer dashboard with real incident data; configured task view cards, removed unwanted action buttons, and added swipe-back navigation component.
Oct 21, 2025	4	Developed dynamic dashboard layouts for Resident and Volunteer roles; added incident cards and status filters.
Oct 22, 2025	6	Implemented nested logic for navigation and api integration based on screens. linked different screen together to build logical flow.
Oct 23, 2025	4.5	Debugged frontend-backend integration for assigned incident retrieval, ensuring dispatch response populates correct reporter and volunteer fields. Rebased main branch, reviewed Harleen's PR.

Description of Work done

Since the previous report, major progress was made on both frontend and backend integration for the CERA application.

Backend

- **Volunteer Assignment System:** Implemented nested assignedVolunteers array with volunteer ObjectId, status, and timestamps. Added controller logic for dispatch and automatic population of reporter info.

- **Middleware Fix:** Refactored verifyToken to attach consistent _id property across all routes, ensuring seamless authentication for /me, /assigned/me, and /dispatch endpoints.
- **API Refactoring:** Improved status filter queries (/incidents?status=approved, /incidents/assigned/me), integrated reporter name on incident creation, and updated error handling.
- **Coordinator Routes:** Added approval and decline logic (/users/:id/approve, /users/:id/decline) with middleware-level role verification.

Frontend Development

- **Volunteer Dashboard:** Connected to backend APIs to display live assigned tasks; removed redundant buttons from task cards.
- **Navigation System:** Introduced custom swipe-back component for Expo Router to return to the previous screen instead of logging out.
- **Authentication Sync:** Updated login screen to store JWT and user details in Encourage and context; added push-token sync with /users/me/push-token.
- **UI Polish:** Refined card layouts, error alerts, and dark/light theme consistency for tasks and incident screens.
- **Documentation:** Updated architecture and Kanban board to include Sprint 3 tasks and epics for Incident Reporting & Volunteer Dispatch.

Testing and Bug Fixes

- Verified middleware flow with Postman and Expo simulator.
- Solved token mismatch between frontend and backend after login.
- Confirmed correct data population of reporterName and volunteer IDs in MongoDB.

Challenges Faced

- **Problem:** Assigned incidents were not appearing in /assigned/me.
Solution: Middleware was sending id instead of _id; refactor fixed the issue.
- **Problem:** Swipe-back gesture caused logout.
Solution: Isolated AuthContext logout trigger and implemented custom BackSwipeHandler.

- **Problem:** Dispatch API failed to populate reporter details.
Solution: Used `populate("reporter", "name email role")` and explicit `reporterName` field on incident creation.

Repo Check-In

Checked-in folders and files since Report 2:

1. Backend:

- `/backend/routes/incidents.js` – added dispatch logic, volunteer assignment, and assigned/me route.
- `/backend/middleware/authMiddleware.js` – rewritten with consistent `_id` field and JWT verification.
- `/backend/models/Incident.js` – updated schema with `reporterName` and `assignedVolunteers` array.

2. Frontend (Expo):

- `/app/screens/TasksScreen.js` – connected to `/incidents/assigned/me` API.
- `/app/components/BackSwipeHandler.js` – new component for custom navigation gesture.
- `/app/context/AuthContext.js` – added state sync and token loading logic.
- `/app/screens/LoginScreen.js` – updated to persist token and trigger push-token API call.

Kanban Board

- **Epic 2:** Basic App & backend skeleton → Task completed (*Show user Dashboard based on the role*).
- **Epic 3:** Registration, Login, Role-Based Access → Task completed (*Role-Based Access*).
- **Epic 4:** Incident Reporting & Volunteer Dispatch → Tasks completed (Not able to fetch nearby incidents after merging to develop, FIX Volunteer Dispatch Screen).
- **Epic 4:** Incident Reporting & Volunteer Dispatch → Tasks in Review/Test (*Incident Reporting & Volunteer Dispatch*).



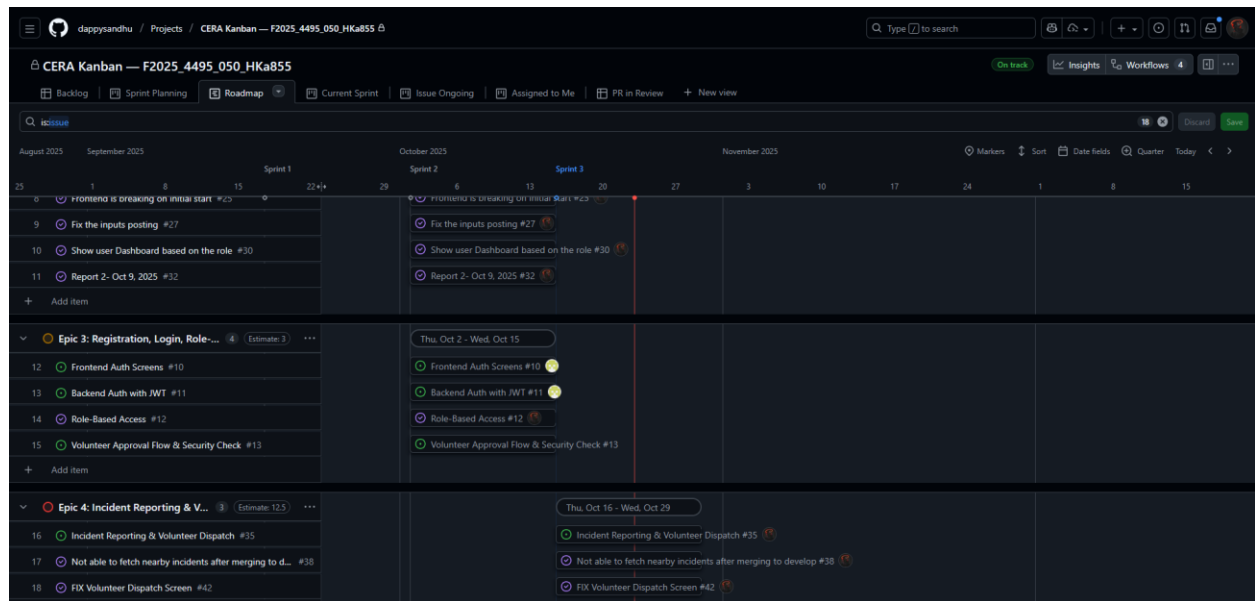


Figure 3: Project Roadmap.

Screens Implemented according to above tasks:

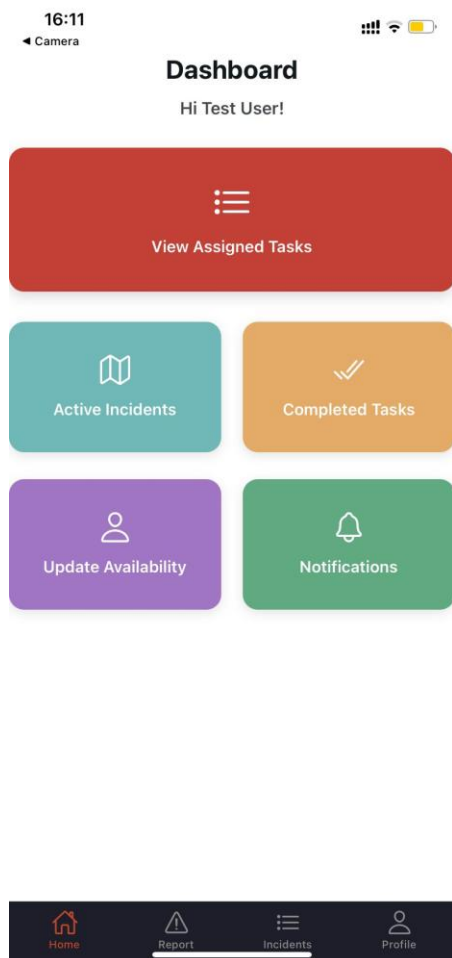


Figure 4(Dashboard)

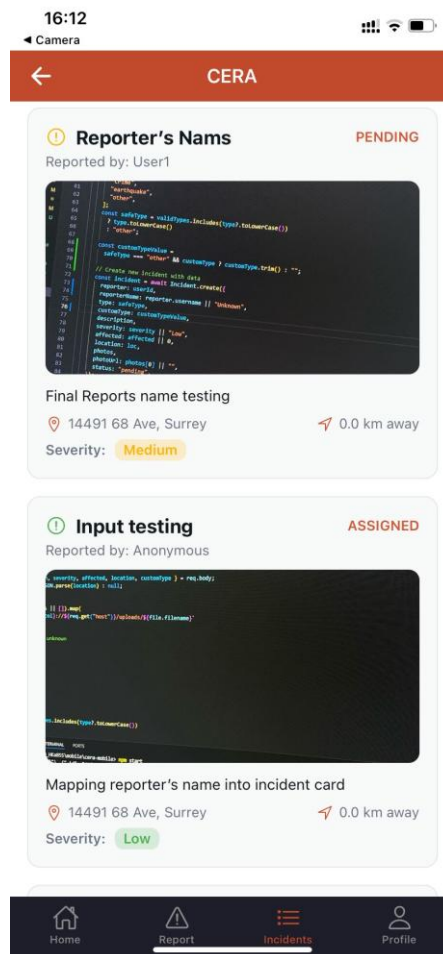


Figure 5(Nearby Incidents)

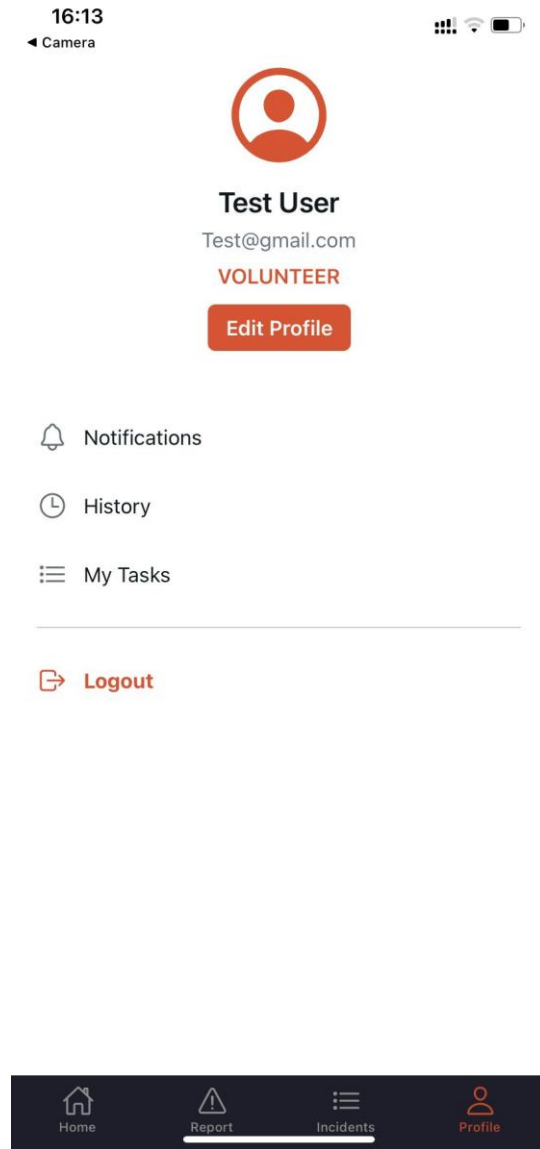


Figure 6(Profile)

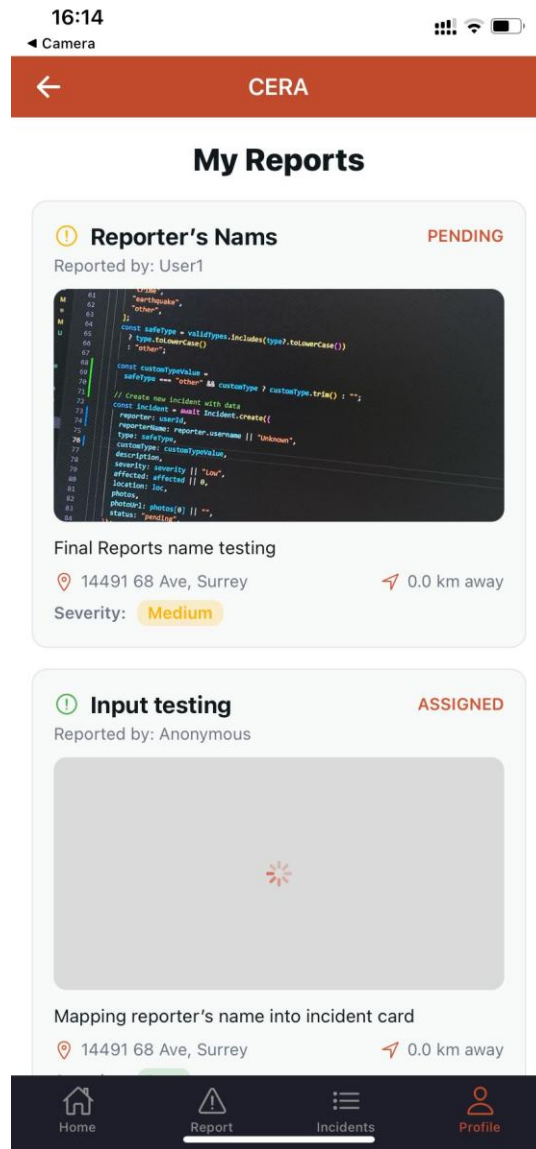


Figure 6.1(My Reports (History))

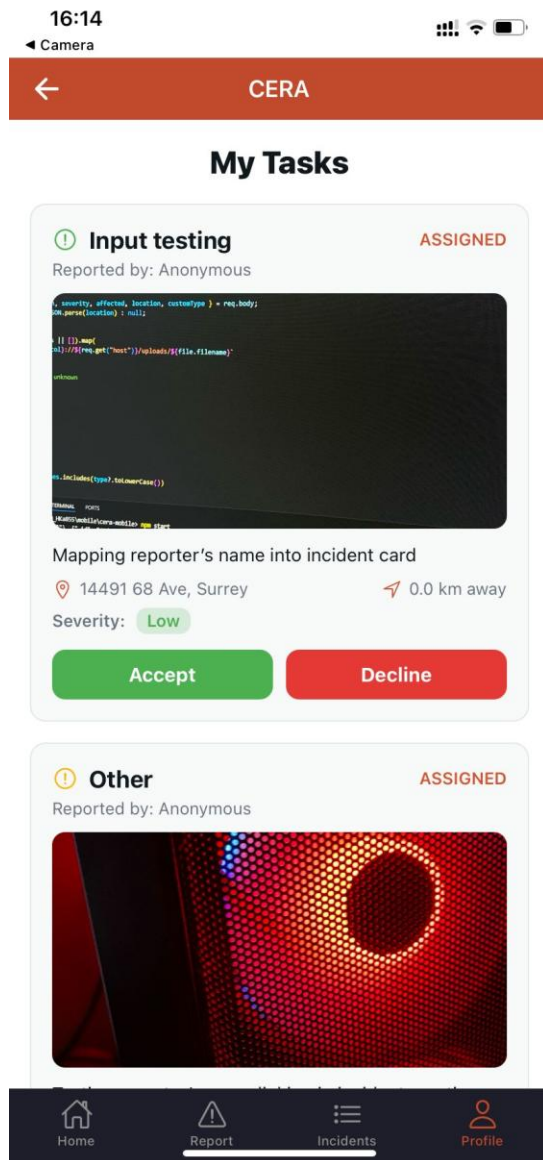


Figure 6.2(My Tasks)

Coordinators Login:

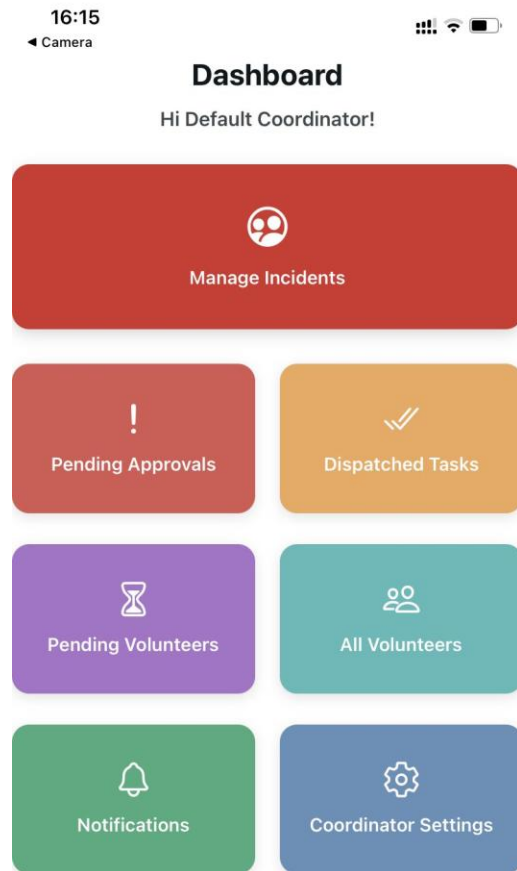


Figure 7(Dashboard)

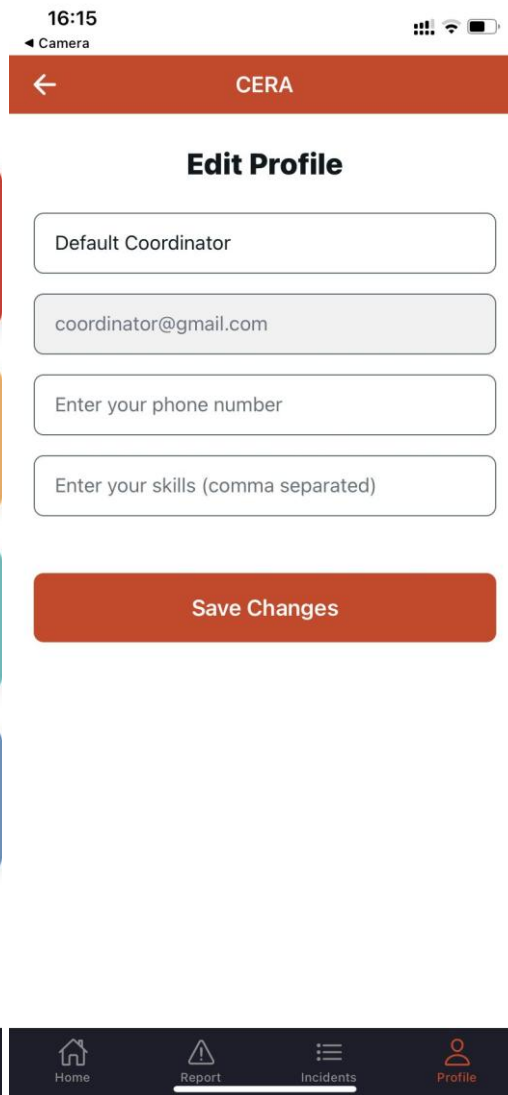


Figure 8(Edit Profile)

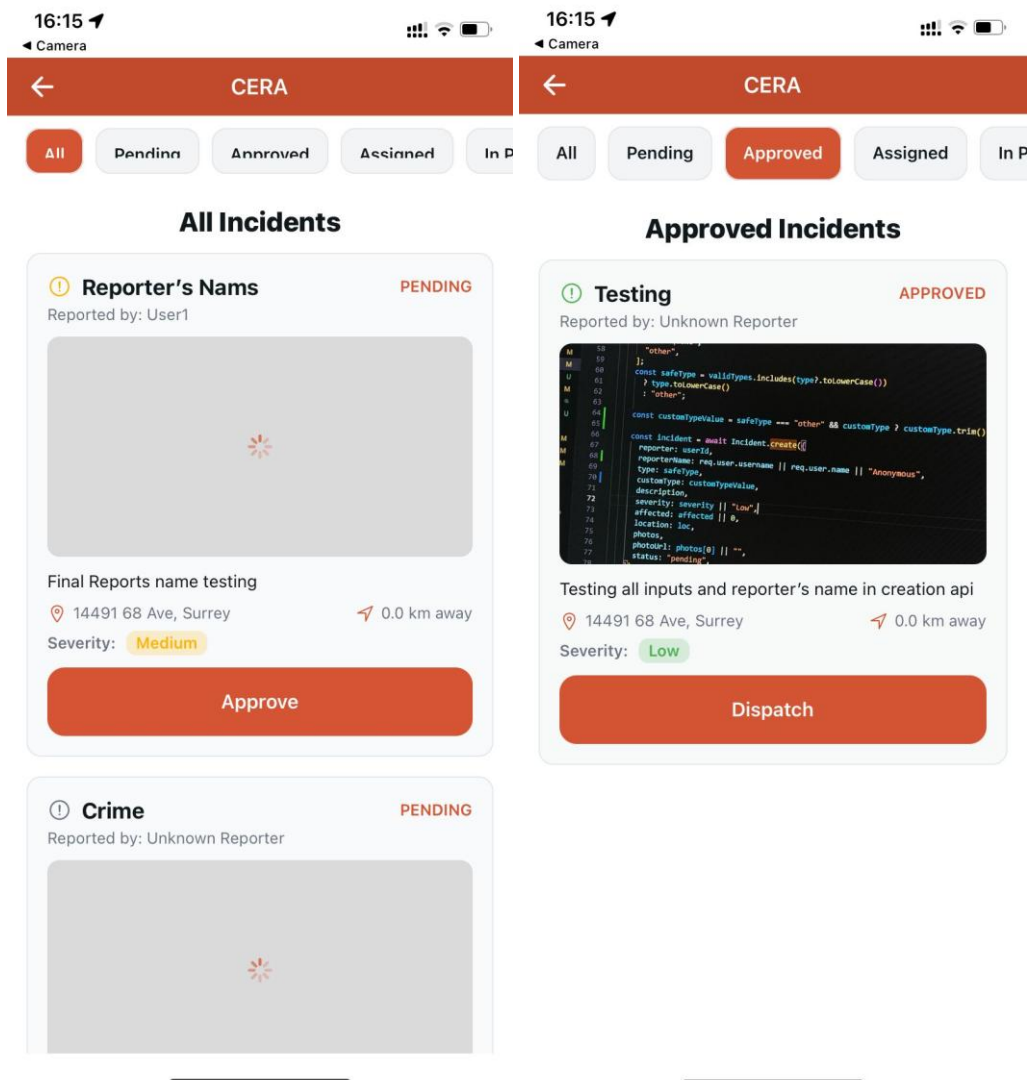


Figure 9(Incidents to Approve)

Figure 10(Incidents Approved)

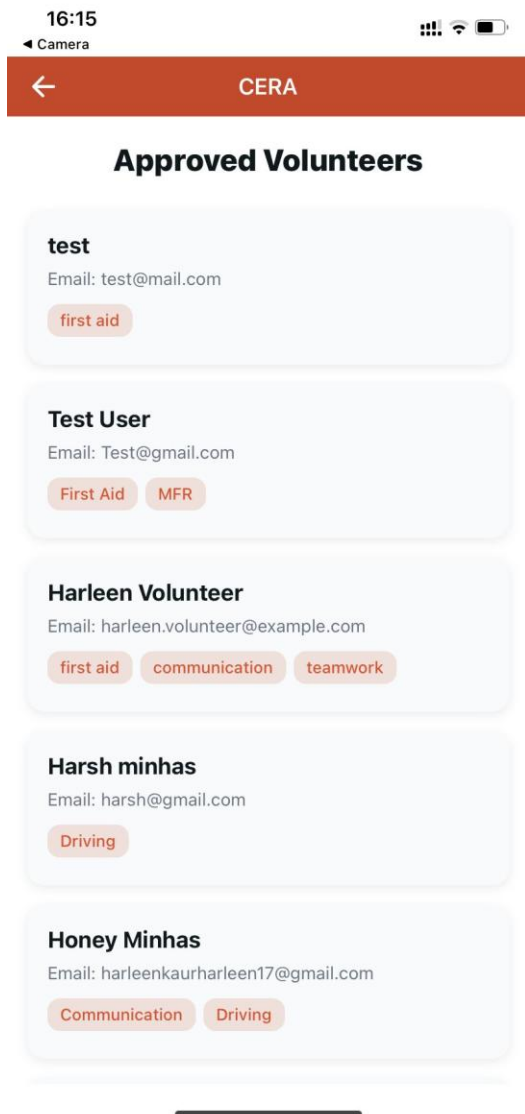


Figure 11(Approved Volunteers)

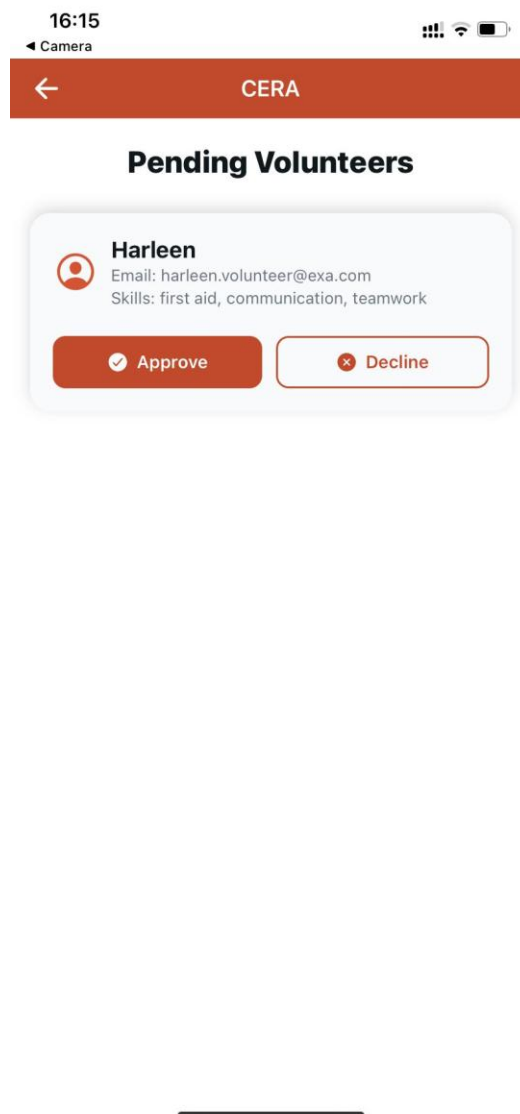


Figure 12(Volunteers to Approve)

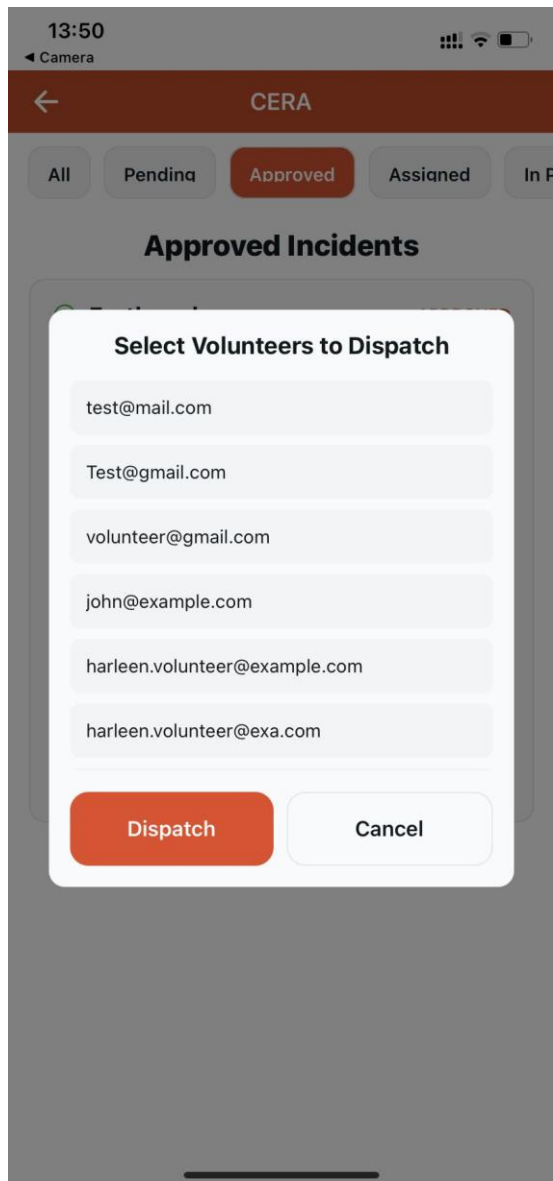


Figure 13(Dispatch Volunteers)